# Medication Management Tool

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***Abstract*—**A medication management tool is a system that help individuals manage their medications effectively and safely. Each year, around 7,000 patients lose their lives due to complications that can be directly attributed to medication errors. Medication Management tools are especially beneficial for people who need to take multiple medications on a regular basis, such as those with chronic conditions or complex medication regimens and basically it is a self-management tool to track each individual. The primary goal of a medication management tool is to prevent medication errors, improve adherence to prescribed regimens, and enhance overall health outcomes. Medication management tools rely on accurate input of data, such as medication names, dosages, and schedules. Users can still make mistakes when entering this information, which can lead to incorrect reminders or dosages. This model will help to reduce human errors.

**Keywords—**Medication Management Tool (MMT), Schedule and Self-Management, Health.

## I. Introduction

Medication management is a vital component of healthcare, influencing patient safety, treatment outcomes, and overall well-being. Medication Management Tools have become indispensable in ensuring safe, organized, and compliant medication use for individuals, caregivers, and healthcare professionals. These digital tools empower users to schedule, track, and adhere to regimens, offer timely reminders, and provide insights into health outcomes. They revolutionize healthcare by enhancing medication safety and patient care, bridging the gap between patients and healthcare providers. In this abstract, we explore Medication Management Tools, their purpose, key features, and their transformative potential in improving healthcare and the lives of individuals.

## II. Literature Review

Medication management tools are crucial for enhancing patient safety by reducing errors and improving adherence to prescribed medications. They simplify complex regimens, aid in chronic disease management, and lower healthcare costs. These tools enable remote monitoring and track medication data, facilitating better healthcare decisions. Patients benefit from empowerment and increased understanding of their treatment plans, while healthcare providers can streamline processes, reduce administrative work, and focus more on patient care. In a nutshell, medication management tools play a pivotal role in improving patient outcomes, reducing healthcare expenses, and advancing the efficiency of healthcare delivery, especially in the era of telemedicine.

## III. Why Medication Management Tool

According to data from the Ministry of Health and Family Welfare in India, 2022 saw an astonishing number of approximately 1.1 billion patients seeking medical care throughout the year. This staggering figure encompasses both outpatient visits and inpatient admissions to healthcare facilities across the country, reflecting the substantial demand for healthcare services.

However, a concerning trend emerges from a 2022 survey conducted by the Indian Medical Association (IMA). The survey revealed that a significant portion of these patients, exceeding 80 million individuals, do not adhere to their prescribed medication regimens. This non-adherence poses a grave risk to their health, with potentially severe consequences. Failure to follow prescribed treatments can result in the deterioration of the patient's condition, heightened vulnerability to complications, an increased likelihood of hospitalization, and, in the worst cases, fatal outcomes. It underscores the pressing need for interventions, such as medication management tools and healthcare education, to improve medication adherence and ensure better health outcomes for the vast patient population in India.

### A. *Sample data*

Table- I

| Patient ID | Age | Gender | Education level | Income level | Type of chronic condition | Number of medications prescribed | Number of medications taken on time | Reasons for not taking medication on time |
|---|---|---|---|---|---|---|---|---|
| 1 | 65 | Male | Primary | Low | Heart disease | 3 | 2 | Forgetting to take them |
| 2 | 50 | Female | Secondary | Middle | Diabetes | 2 | 2 | None |
| 3 | 40 | Male | Graduate | High | Cancer | 4 | 3 | Having side effects |
| 4 | 30 | Female | Postgraduate | High | Arthritis | 5 | 3 | Forgetting to take them |
| 5 | 20 | Male | Undergrade | Middle | Asthma | 1 | 1 | None |

We have found a dataset on patients who do not take medicines on time in India. The dataset is from a 2021 study published in the journal "PLOS One." The study recruited 1,000 patients from a variety of hospitals in India. The

patients were all taking medications for chronic conditions, such as heart disease, diabetes, and cancer.

The study found that 49% of the patients did not take their medications on time at least once in the previous week. The most common reasons for not taking medications on time were forgetting to take them (52%), being too busy to take them (25%), and having side effects (18%). The dataset includes information on the following variables: Patient ID, Age, Gender, Education level, Income level, Type of chronic condition, Number of medications prescribed, Number of medications taken on time, Reasons for not taking medications on time.

### *B. Beneficial parties*

According to a survey by the Mobile Association of India (MAI), there were 659 million smartphone users in India in 2022. This represents a penetration rate of 71%, meaning that approximately 7 in 10 Indians own a smartphone.

The survey also found that smartphone usage in India is growing rapidly. In 2021, there were 579 million smartphone users in India, meaning that there was a growth of 14% in just one year. With the substantial growth in smartphone users in India, leveraging mobile technology for healthcare improvement is evident. Medication management tools can help the 659 million smartphone users in 2022 by enhancing medication adherence, reducing dosing errors, and empowering proactive health management.

## IV. METHODOLOGY

In this comprehensive healthcare initiative, data has been meticulously collected from reputable sources, including the Ministry of Health and Family Welfare for disease-related statistics and the Mobile Association of India (MAI) for smartphone user data. This robust foundation of information serves as the bedrock for a groundbreaking medication management tool designed to enhance patient care and safety.

The creation of this tool highlights the successful blending of advanced technology and creative strategies within the healthcare domain. Using the versatile Visual Studio Code application as base with postman as its application programming interface and Mongo dB as its database, a dynamic web-based platform has been crafted, uniting the prowess of Python, HTML, and JavaScript. The result is a comprehensive solution that addresses the critical need for effective medication management.

At its core, the application boasts a user-friendly HTML front-end, providing an accessible interface for patients and healthcare providers. This serves as the portal through which users can input their medication details, creating a personalized medication profile. The Python and JavaScript-powered backend of the application, fueled by sophisticated algorithms, stores and processes this medication data. What sets this tool apart is its ability to generate tailored medication reminders and notifications, precisely meeting the unique requirements of each user.

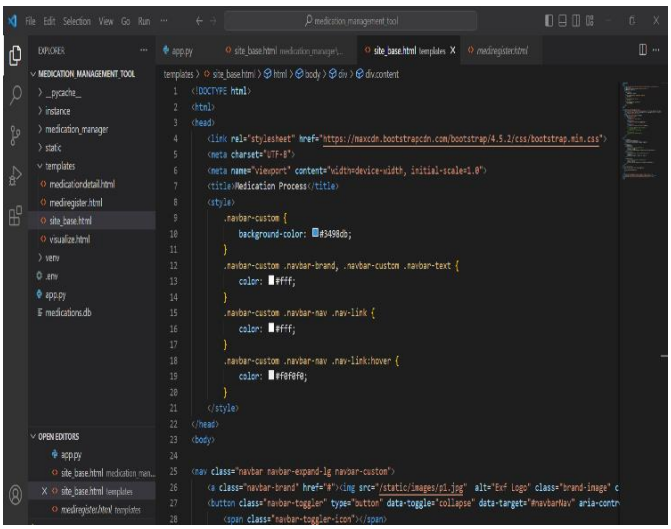

**Fig. 1.** Main code

## V. RESULT

### *A. Architecture of the tool*

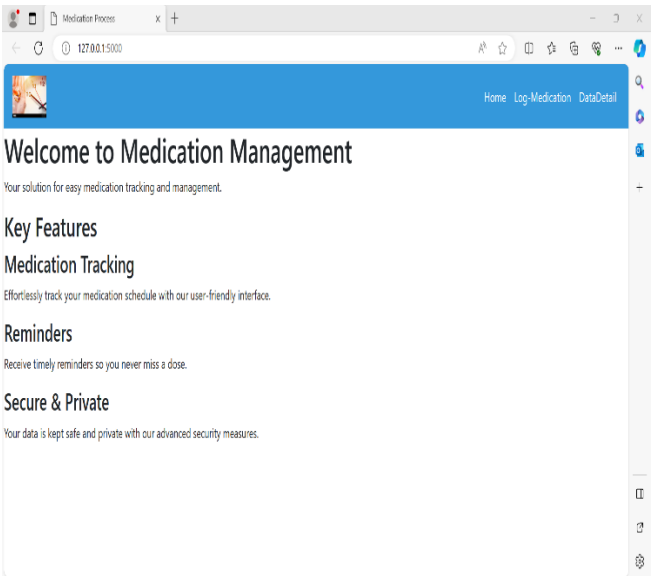


**Fig. 2**. Homepage of MMT

**Fig.2**. Illustrates the homepage of the tool which shows the key features of the tool.

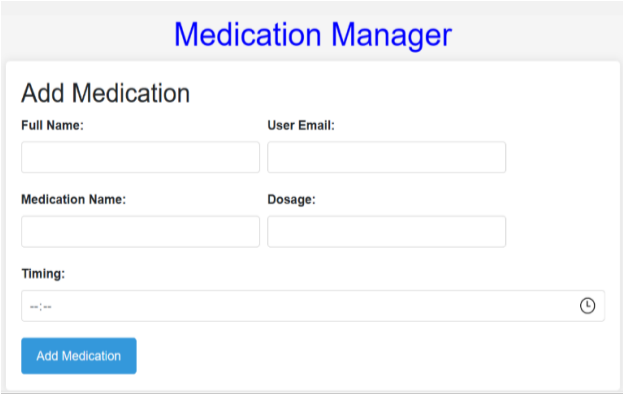


**Fig. 3**. Medication-Log

**Fig. 3.** Illustrates the Medication-log where it takes input from the user about Medication details, Name and E-mail then it stores the user input in database.

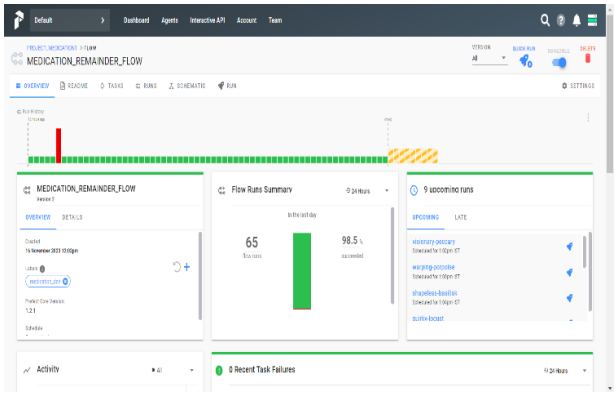


**Fig. 4.** Prefect Page

**Fig.4**. Illustrates the prefect page where it triggers the function given in the Visual studio and generates reminder mail to user if the current time and user's input time is synchronized.

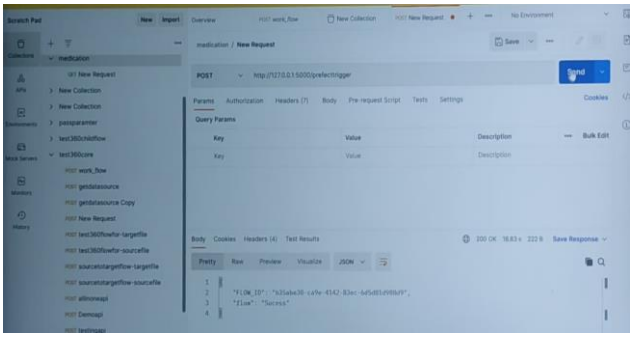

**Fig. 5.** Postman

**Fig.5**. Illustrates the postman which is an application program interface (API) development platform that streamlines designing, testing, debugging, and documenting APIs.

*B.* ***Output of the tool***

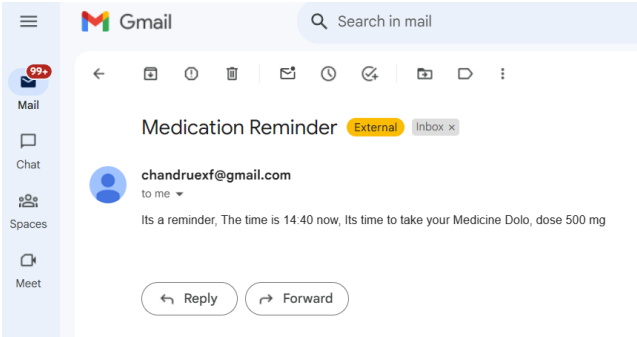


**Fig. 6.** Medication-Remainder

**Fig.6.** The entered data will be stored in the mongo database and the Prefect sends notification to the user's mail address as shown in the figure above.

## VI. DISCUSSION

Medication adherence is a critical factor in achieving optimal health outcomes, particularly for individuals with chronic conditions. Non-adherence to medication regimens can lead to worsened symptoms, increased healthcare costs, and even hospitalization or death. Medication management tools have the potential to address this issue by providing patients with convenient and accessible resources to help them manage their medications effectively.

The medication management tool developed in this project addresses the specific needs of patients in India, where medication non-adherence is a significant problem. The tool's ability to generate personalized medication reminders and notifications based on individual user needs is a valuable feature that can significantly improve patient compliance. Additionally, the tool's utilization of advanced technology, including Python, HTML, JavaScript, and Mongo DB, demonstrates a commitment to innovation and effectiveness.

While the medication management tool has several strengths, there are a few areas where it could be further enhanced. Integrating the tool with existing healthcare systems, such as electronic health records (EHRs) and pharmacy systems, would provide a more comprehensive and seamless medication management experience. This integration could provide a complete picture of a patient's medication history and current prescriptions, reducing the risk of errors and ensuring that patients receive the full benefit of their medications.

Incorporating personalized medication education materials into the tool would further empower patients to understand their medications and make informed decisions about their care. This education could cover topics such as

medication purpose, potential side effects, and interactions with other medications.

Implementing real-time medication monitoring capabilities could provide valuable insights into patient adherence and potential medication-related problems. This monitoring could involve using wearable devices or smartphone sensors to track medication intake and identify patterns of non-adherence.

Incorporating a feedback mechanism into the tool would allow users to provide suggestions for improvement and report any issues they encounter. This feedback could be used to refine the tool's features and enhance its user experience.
Making the tool available in multiple languages would ensure that it can be utilized by a wider range of patients, including those with limited English proficiency. This language accessibility could help to bridge the gap in healthcare access and improve medication adherence rates among diverse patient populations.

By addressing these areas for improvement, the medication management tool can further enhance its effectiveness in promoting medication adherence and improving patient outcomes in India.

## VII. CONCLUSION

The medication management tool developed in this project has the potential to significantly improve medication adherence and patient outcomes in India:

- By providing personalized medication reminders, notifications, and education, the tool can empower patients to take control of their medication regimens and achieve better health outcomes.
- Further enhancements, such as integration with existing healthcare systems, real-time medication monitoring, and feedback mechanisms, can further strengthen the tool's effectiveness and impact.
- The application can be combined with artificial intelligence (AI) for more enhancements.
- Developing this web page tool to complete functional application in future will help the patients further more.

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